

Transmission Spectra in EXHALE_transit.py:

$H\alpha/H\beta$ from the Non-LTE $n = 2$ Population,

and the Metal Resonance Doublets

Contents

1. Overview	2
2. The $n = 2$ hydrogen population (Christie et al. 2013)	2
2.1 Rate coefficients	3
2.2 Ly α pumping and the $J_{Ly\alpha}$ input	3
2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)	4
3. $H\alpha$ and $H\beta$ optical depth and transmission	5
4. Implementation in EXHALE_transit.py	5
4.1 Outputs	6
4.2 Which state the spectrum is built from	7
4.3 Environment variables	7
5. What the spectrum is built on: row validity, the census, and the metadata block	8
5.1 Reading the profile's own statement of validity	8
5.2 The census: which rows the chords sample	9
5.3 The metadata block on every saved curve	10
6. Validation	11
7. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)	11
7.1 Atomic data and implementation	11
7.2 Example spectra	12
8. Comparison with an observed line profile	12
8.1 The model line is matched to the observed width	12
8.2 The equivalent-width operator	14
8.3 The line-of-sight shift is for figures only	15
9. Caveats	15

1. Overview

EXHALE_transit.py (the Transmission Probability Model shipped with EXHALE) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2³S triplet (10830 Å) and the H I / D Ly α doublet (1215.67 Å).

This memo documents the addition of the **H α (6562.8 Å, $n = 2 \rightarrow n = 3$)** transmission spectrum. Unlike Ly α , which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, H α absorbs out of the $n = 2$ level, whose population must be computed from a non-LTE balance. We follow **Christie, Arras & Li (2013, ApJ 772, 144)**, including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and **Ly α radiative pumping** ($1s \leftrightarrow 2p$), with the rate coefficients of EXHALE's own $n = 2$ model. The Ly α mean intensity $J_{\text{Ly}\alpha}(r)$ is not in the profiles the tool reads (a run with the excited-hydrogen model writes its own field to output/Excited_H.txt, which is not read); it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017), see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether Jlya_file is set. The existing He/Ly α behavior is unchanged.

This memo also documents the **metal resonance doublets** (Mg II h&k, Ca II H&K, Na I D; §7.), which are computed by the same line-of-sight integration and produce the same three-curve spectrum figures as the H/He lines, and the convention by which a model line is compared with an observed transit profile: the width-matching velocity convolution, the equivalent-width operator and the line-of-sight shift used only in figures (§8.).

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The 2s and 2p states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.2649 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.22461 \text{ s}^{-1}$, Drake 1986, eq. 27). With $n_{1s} = n_{\text{HI}}$, n_{HII} , n_{HeII} , n_{HeIII} , n_e and T taken from the EXHALE profiles, the rate-equilibrium equations are those of Christie et al. (Eqs. 12–13) as the solver writes them (excited_hydrogen.f90, n2_rate_matrix):

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + C_{2p \rightarrow 1s} n_e + M_{2p2s} + \Gamma_{2p}] n_{2p} - M_{2s2p} n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e n_{\text{HII}} + \frac{3}{4} S_{\text{chem}}, \quad (1)$$

$$[C_{2s \rightarrow 1s} n_e + M_{2s2p} + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - M_{2p2s} n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e n_{\text{HII}} + \frac{1}{4} S_{\text{chem}}, \quad (2)$$

where the recombination source is $\alpha n_e n_{\text{HII}}$ (the electron recombines onto a proton; n_e and n_{HII} differ at the base, where helium and metals supply the electrons), $M_{2s2p} [\text{s}^{-1}]$ is the ℓ -mixing rate by electrons and ions, $M_{2p2s} = (g_{2s}/g_{2p}) M_{2s2p}$, and S_{chem} is the chemical $n = 2$ production of a molecular run, the dissociative recombination of H_2^+ and HeH^+ ($k_5 n_{\text{H}_2^+} n_e + k_{16} n_{\text{HeH}^+} n_e$, Koskinen et al. 2022 R5 and R16), shared between 2s and 2p by statistical weight. This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -M_{2s2p} \\ -M_{2p2s} & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The $2s$ and $2p$ densities are carried separately into the Balmer opacity, whose sub-level oscillator strengths differ.

2.1 Rate coefficients

Every coefficient is the one the solver's level balance uses, transcribed in `exhale_transit_lib.py` from `hydrogen_n2_rates.f90` and `Cool_coeff.f90` (the Fortran name is given at each function, and the two agree to 3×10^{-13} relative over 10^3 – 10^5 K, the ℓ -mixing rates also over $n = 1$ – 10^{15} cm $^{-3}$, measured against the production objects):

Coefficient	Source and form
α_B	the case-B coefficient of the ionization balance: Badnell (2006, 2023 update) radiative recombination minus the ground capture of the Milne relation on the hydrogenic cross section (or the Koskinen et al. 2022 or Hui & Gnedin 1997 fit, as <code>Atomic_rate_set</code> and <code>Legacy_HHe_rates</code> of the run's <code>input.inp</code> select)
α_{2s}, α_{2p}	$f_{2s}(T)\alpha_B$ and $\alpha_B - \alpha_{2s}$, with f_{2s} the case-B fraction of Pengelly (1964, Table I); 0.322 at 10^4 K
$C_{1s \rightarrow 2s}, C_{1s \rightarrow 2p}$	$C \Upsilon(T) e^{-E_{12}/kT} / (2\sqrt{T})$, $C = h^2 / [(2\pi m_e)^{3/2} k^{1/2}]$, the CHIANTI v11.0.2 h_1 effective collision strengths (the rates of the HI cooling)
$C_{2s \rightarrow 1s}, C_{2p \rightarrow 1s}$	their detailed-balance reverses, Boltzmann factor cancelled analytically
M_{2s2p}	$n_e q^e + n_{\text{HII}} q^p + n_{\text{HeII}} q^{\text{He}^+} + n_{\text{HeIII}} q^{\text{He}^{2+}}$: electrons from Seaton (1955, eq. 55, approximation V; 5.78×10^{-5} at 10^4 K), ions from the impact-parameter cross section of Pengelly & Seaton (1964) with their cut-off (protons 4.8×10^{-4} at 10^4 K)

The rates of Christie et al. (2013, Table 2) used here before 2026-09-25 differed in four places: $A_{2s \rightarrow 1s} = 8.26$ s $^{-1}$; the Draine case-B fit with a quadratic f_{2s} ; power-law collision rates; and the electron ℓ -mixing rate of their R5 (5.35×10^{-6} at 10^4 K) with no ion term. Only the last moves the populations measurably: on the HD 209458 b _adv profile the proton term brings n_{2s}/n_2 from 0.53–0.69 down to 0.26–0.34 between 1.3 and $3 R_p$, the peak H α depth falls by 1.7 per cent and H β is unchanged; on the solved wasp_full state H α rises by 0.14 per cent (measured, `md/stale_texts_20260925.md`).

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p}) B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units erg s $^{-1}$ cm $^{-2}$ Hz $^{-1}$ sr $^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s $^{-1}$.

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If `Jlya_file` points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1 r/R_p (same radial coordinate as the EXHALE output)
column 2 $J_{\text{Ly}\alpha}$ [$\text{erg s}^{-1} \text{cm}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$]

linearly interpolated onto the EXHALE grid and clamped to the endpoints. A template ships as `inputdata/Jlya.txt.example`.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere ($2.6 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, EXHALE_transit.py computes the absorbed LyC flux from the EXHALE inputs:

$$F_{\text{LyC}} = \xi \underbrace{\frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{HI}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6 \text{ eV}$) and the orbital distance a are read from `input.inp`, N_{HI} is the vertical neutral-hydrogen column from the EXHALE profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18} \text{ cm}^2$. The factor ξ is the day-night / 2D flux dilution, taken to match ATES's "2D approximate method" (`Rate/2` $\rightarrow$ 0.5, `Rate/4` $\rightarrow$ 0.25, else 1; read from `input.inp`, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ for the HD 209458b input). A fixed value can be forced via `F_LyC_override` and the dilution via `xi_override`.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply `Jlya_file`.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4 \text{ eV}$, $\lambda < 3647 \text{ \AA}$); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The solver integrates the Balmer band of the run's own spectrum type, but neither that spectrum nor the rate is in the profiles this tool reads, so here the field is a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\nu_1} \frac{F_\nu}{h\nu} \sigma_2(\nu) d\nu, \quad F_\nu = \pi B_\nu(T_\star) \left(\frac{R_\star}{a}\right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu}\right)^3,$$

over the band the solver uses, from the $n = 2$ edge $h\nu_2 = 3.3996 \text{ eV}$ to the H I edge $h\nu_1 = 13.5984 \text{ eV}$ (above it the ground state absorbs the photons), with $\sigma_{2,\text{th}} = 1.4 \times 10^{-17} \text{ cm}^2$ for both levels and $\Gamma_{2s} = \Gamma_{2p} = \Gamma_2$. The solver states no

source for $\sigma_{2,\text{th}}$; it is, to its two figures, the $n = 2$ threshold value of the Kramers cross section with the bound-free Gaunt factor of Seaton (1959, MNRAS 119, 81, eqs. 3 and 10), $1.378 \times 10^{-17} \text{ cm}^2$, and the exact hydrogenic threshold values are $1.478 \times 10^{-17} (2s)$ and $1.355 \times 10^{-17} (2p)$ cm^2 , statistical mean $1.386 \times 10^{-17} \text{ cm}^2$ (derived by direct integration of the bound-free dipole matrix elements). This reproduces Huang et al.'s Table 2 values ($\sim 20\text{--}26 \text{ s}^{-1}$) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. $\sim 140 \text{ s}^{-1}$ for HD 209458's $T_\star \simeq 6065 \text{ K}$), which suppresses n_2 and hence the Balmer-line depth. Setting $T_\star \leq 0$ falls back to the manual constants $\text{Gamma}_{2s}/\text{Gamma}_{2p}$ (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS integration as Ly α . Along a chord at impact parameter b ,

$$\tau_\nu(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8 \text{ \AA}$ (air), $f_{23} = 0.6407$, $A_{3\rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35 \text{ \AA}$ ($f_{24} = 0.119$, $A_{4\rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, H $\alpha \simeq 3.2\%$ and H $\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_\star, T_\star$ (see the comparison notebook).

4. Implementation in EXHALE_transit.py

System parameters. The planet and the star are taken from the run directory, matched by label exactly as `input_read.f90` does. Besides $R_p, M_p, T_{\text{eq}}, a, M_\star, L_{\text{EUV}}$ and the 2-D approximate method, this includes the stellar radius (`Stellar radius [R_sun]`) and effective temperature (`Stellar Teff [K]`), which set the transit normalization $A_\star$, the $R_\star/R_p$ cap on the absorbing annulus, and the diluted-blackbody Balmer continuum of §2.3. The planet spin period has no `input.inp` key: a close-in giant is tidally locked, so it defaults to the Keplerian orbital period $P_{\text{orb}} = 2\pi\sqrt{a^3/[G(M_\star + M_p)]}$. Each of the three is resolved as `EXHALE_TRANSIT_RSTAR_RSUN / _TSTAR / _ROTP` environment override `> ./input.inp > built-in default`, and the resolved values and their sources are printed at startup.

EXHALE_resolved.out comes first (2026-08-22). $R_p, M_p, T_{\text{eq}}, a, M_\star$ and He/H are read from `EXHALE_resolved.out` beside `input.inp` whenever that file exists, and only from `input.inp` when it does not. The wind solver writes it (`write_resolved_config` in `write_setup_report.f90`) after a `base.inp` handoff or a lower-atmosphere profile has moved the base, so it is the geometry the wind actually used, while `input.inp` alone is stale in exactly that case.

read_input_params says at startup which of the two it took, and a file that is present but unreadable raises rather than falling back silently. The stellar radius and T_{eff} are not in that file and keep the precedence above.

- `n2_populations(T, n1s, nHII, nHeII, nHeIII, ne, Jlya, G2s, G2p, nH2p, nHeHp, rate_set)` (exhale_transit_lib.py) solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} . The Ly α -pump constants $B_{1s \rightarrow 2p}$, $B_{2p \rightarrow 1s}$ are precomputed.
- $J_{\text{Ly}\alpha}$ is read from `Jlya_file` if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile `data_n2` (in m^{-3}), exactly like `data_nHI`.
- A new wavelength grid `l_onda_Ha` and an H α LOS loop fill `exp_tau_Ha`; the disk average, instrument convolution (`Instr_res_Ha`) and rotation convolution produce the plotted curves.
- The H α branch always runs (`do_Ha=True`); with the default `Jlya_file=''` it uses the Huang et al. (2017) estimate for $J_{\text{Ly}\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).
- **Compatibility fix.** `Ion_species.txt` in EXHALE carries trace-metal columns, so the reader was changed to `usecols=range(7)` to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs. The column count has since grown: `write_output.f90` writes r , the six H/He species (HI HII HeI HeII HeIII HeITR) and all `n_mion = 27` metal-ion stages, i.e. **34** columns, and **38** on a molecular run, which appends H2 H2p H3p HeHp; the metal columns are written whether or not metals are on. The `usecols=range(7)` slice is unaffected (the first seven columns are still r plus the six H/He species), and anything past them is found through the file's own `# columns` schema header.

4.1 Outputs

Every line carries one key (He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI, and OI for the OI 1302 triplet when the run wrote `OI_levels_adv.txt`), and both products of a line are named from that key and land in the run directory:

Product	Name	Default
model curve	<path>/tpm_<line>.txt	always written
figure	<path>/<prefix><line>.png (+ .pdf)	off; set EXHALE_TRANSIT_FIG_PREFIX

A curve file has the columns `lambda[A]`, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$. The set follows the run: a metals-off run writes no metal curves. `EXHALE_TRANSIT_SAVE_PREFIX` decorates the curve name exactly as the figure prefix decorates the figure name, and both are resolved against the run directory, so an absolute prefix writes elsewhere and nothing depends on the working directory. Every curve file also opens with a block of `#` comment lines stating what the spectrum was built on: the input profile and its validity schema, the census of the rows the chords sampled, and the overrides in effect (§5.3). They are comments, so the four numerical columns are what they always were.

Two further files carry the same metadata block. `tpm_He10830_metrics.txt` holds the He 10830 line metrics of the three-Gaussian extractor `he_line_metrics.fit_metrics` (the model of Cherubim et al. 2026: three Gaussians at the fixed triplet positions with one shared width and one shared shift), fitted in the air frame to the excess ($\max T_\lambda -$

T_λ)/ $\max T_\lambda$ of the planet-rotation and instrument-convolved curve (column 4; the instrument-only column 3 until 2026-10-06): red_depth, blue_depth, red_blue, fwhm_A (of the blended red pair), shift_A, sigma_A. A failed fit is reported and is not fatal. tpm_OI_band_depths.txt holds the OI band depths and the published HD 209458 b values they are compared with.

4.2 Which state the spectrum is built from

A run can hold two states of one column, and the tool reads one pair of files from one of them, never the temperature of one and the composition of the other (transit_state_files, exhale_transit_lib.py). EXHALE_TRANSIT_STATE selects it:

- adv (default; also post_process, corrected): output/Hydro_ioniz_adv.txt and output/Ion_species_adv.txt, the advection-corrected profile, a post-process of the state that is an independent discretization of the same column;
- solution (also solved, state): output/Hydro_ioniz.txt and output/Ion_species.txt, the state the wind solver produced and the certification judged. With Ionization transport: True its composition is the transported partition of the ionization stages.

A stationary solve writes no _adv pair; the evaluation of a solved state (Restart intent: stationary evaluate) runs the post-process and writes it. A selection whose pair is missing is refused by name (the script exits with the message), and the other state is never read in its place. The derived-state refusal of §5.1 (exit status 7) is applied to the adv selection only. Every saved file records the selection (state_selection), the two file paths, the md5 and size of each file read (state_file_identity), the provenance lines copied from the state's own header, and the largest relative distance from the run's other state over the rows both carry, in T , n_{HI} and $n_{\text{He}(2^3\text{S})}$ (state_difference), or a line saying that distance is UNKNOWN when the other state is absent or on another grid.

4.3 Environment variables

Every override is read as EXHALE_TRANSIT_<name>, with the old TPM_<name> still accepted (_tenv in exhale_transit_lib.py); the overrides in effect are written into every product. The full set that EXHALE_transit.py and exhale_transit_lib.py read:

EXHALE_TRANSIT_...	default	meaning
PATH	.	run directory (input.inp, output/); products land there
STATE	adv	state selection, §4.2
SAVE_PREFIX, FIG_PREFIX	empty	curve-name prefix; figure prefix (figures written only when set)
DIAGNOSTIC	0	1 makes curves from a rejected or record-less adv profile, as diagnostic data
RES_HETR	8×10^4	resolving power, He 10830 (CARMENES; LHS 1140 b uses 68 000, LHS1140b/winered_hires_y.sh)
RES_HI	5×10^4	Ly α
RES_HA, RES_HB	1.15×10^5	H α , H β
RES_MGII	3×10^4	Mg II
RES_CAI, RES_NAI	RES_HA	Ca II, Na I
RES_OI	10^3	O I 1302
HE_LMIN, HE_LMAX, HE_N	10828.2, 10831.2, 201	He window [$\text{\AA}$, air] and its sampling (widened automatically for a fast wind, unless either bound is set)
RSTAR_RSUN, TSTAR, ROTP	input.inp	stellar radius, T_{eff} , planet spin period (default P_{orb})
ROT_NPHI	64	azimuthal samples of the rotation integral
TURB	0	1 adds the p-winds turbulence $v_{\text{turb}}^2 = (5/6)kT/m$ in quadrature to the thermal width
OI_WEIGHT, OI_STAR_FWHM_A,	star; 0.2;	O I band weight (stellar line profile or flat), its width and
OI_STAR_PEAKS, OI_NSACLE	1.0, 1.5, 1.17; 1.0	peaks, and a level-density scale for curve-of-growth tests

5. What the spectrum is built on: row validity, the census, and the metadata block

By default the spectra are integrated over the *_adv.txt profiles (§4.2), and those profiles do not carry one uniform product: the steady advective correction is applied to some rows and refused on others. Since the profiles began stating that row by row, the transit tool reads the statement, reports what the chords sampled, and writes both into every curve it saves. None of this changes a numerical column of a tpm_*.txt file.

5.1 Reading the profile's own statement of validity

read_adv_validity(exhale_transit_lib.py) decides which of three cases a profile is, by its header and by nothing else. Columns are located by name on the # columns line, so a file that gains or reorders columns is still read correctly.

adv_schema	What the file carries	Row decision
2	adv_T_status, adv_comp_status, adv_mass_row	refused when adv_T_status $\neq 0$ or adv_comp_status = 2
1	one adv_status column	refused when it is nonzero
absent	no schema line and no status column	every row UNKNOWN

Under schema 2 each status field takes one of five values, 0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not_evaluated. The census refuses a row whose temperature field is not zero, or whose composition field is failed: a corrected temperature on a composition whose solve did not converge is not a corrected row (refused_reason is 2 when either field is failed, else the temperature field). A retained composition under a corrected temperature is not refused; the composition field is reported beside the census, because the lower-level density of every line comes from that composition. A refused field of the profile carries the run's own numbers exactly, except the temperature of a failed row of a rejected column energy solve, which is the marching profile of the post-process; such a row exists only in a profile whose record is rejected (below).

A corrected row is a *conditional* correction, not a certified one. The profile states the fraction it is accurate to in the mass flux, `adv_conditional_tol` (currently 10^{-2} , the first-order argument at its declaration), and the measure each row was decided by, `adv_mass_row` = $|R_1|/s_1$, the mass row of the certification's own residual scaled by that row's own largest term. The header's `adv_input_certified` line is a different statement and is kept separate: it is the certification's verdict on the run state, at the much tighter `cert_tol_mass`, not a verdict on the post-processed rows. The reader returns the tolerance, its header text and the measure column, so the spectrum can carry the condition of the rows it stands on.

The derived-state record. Beside the row fields the profile carries one # `adv_derived_state:` line, the record of how the solves that produced the rows ended as a whole (the column energy solve of the transport terms, the chemistry, the outer iteration), read back as the dict `derived_state` and judged by `adv_derived_state_verdict:` rejected, unverified (the best a record says: the outer iteration is a fixed number of passes and the column is solved with lagged coefficients), or unknown when the file has no such line. A rejected or unknown profile gives no spectrum: EXHALE_transit.py exits with status 7, the status the evaluation itself exits with for a rejected product, unless EXHALE_TRANSIT_DIAGNOSTIC=1, which makes the curves as diagnostic data. The solution selection has no such record and is not judged by one.

A legacy profile is read as UNKNOWN and said to be UNKNOWN. It is not read as "corrected": a writer that did not state its refusals says nothing about them, and a census of such a file can only say so. Both the console and the saved metadata print that distinction explicitly, because UNKNOWN is not zero refused.

5.2 The census: which rows the chords sample

A ray at impact parameter b crosses every shell with $r \geq b$, at line-of-sight coordinate $|x| = \sqrt{r^2 - b^2}$, and there is *no upper radial limit*: a shell at $r = 12$ meets the ray at $b = 2$ whether or not the ray grid stops at $b = 10$, so a shell lying outside the stellar disk still absorbs on the rays that cross the disk. One function, `chord_shell_indices`, now serves the census, `resonance_depth` and `resonance_spectrum` alike; two selections would let the diagnostic describe a different atmosphere from the one the spectrum was built from. The earlier census mask imposed an upper limit at the stellar-disk cap R_{ib} and therefore described less material than the spectrum contains. That defect is latent whenever R_{ib} is the outermost grid radius, which both regression profiles are, and real as soon as the domain passes the stellar radius: MEASURED on `wasp_full` read at EXHALE_TRANSIT_RSTAR_RSUN=0.30, the sampled rows go from 419 to 500, 81 of them above the cap.

The console reports, and the metadata block records, how many of the sampled rows are not the steady advective correction, broken down by reason and with the composition field beside it. Then, line by line at eight impact parameters and at its maximum, it reports the share of the line-center optical depth carried by those rows. The share is exact rather than estimated: the optical depth of a ray is the trapezoid sum $\tau(b) = \sum_k w_k I_k$ over the rows the chord crosses, which is additive, so a subset of the rows carries exactly $\sum_{k \in \text{subset}} w_k I_k / \tau(b)$ (refused_line_center_tau_share).

It is a CONTRIBUTION, not an uncertainty. The disk-averaged transmission is an average of $\exp(-\tau)$ and is nonlinear in τ , so a saturated ray moves very little when its optical depth moves, and a share near one does not mean the depth is wrong by a comparable amount. The tool labels it as such in the console and in the file, and it should be quoted that way. On `wasp_full` the rays near $b = 1$ take 99 percent of their Ly α line-center depth from uncorrected rows (MEASURED), which says where the material is, not that the depth is unreliable.

5.3 The metadata block on every saved curve

`transit_metadata_block` writes a block of # comment lines at the head of every `tpm_*.txt` product, the two hand-written products included. Comments only: `np.loadtxt` and any reader that skips comment lines sees exactly the numerical columns it saw before, and the numerical columns were verified bitwise across all ten `wasp_full` and five `mol_base_handoff` products when the block was introduced. The block carries, in order:

- `transit_schema` (currently 1) and a one-line statement that the product is a one-way transmission of the profile named next;
- the input profile's path, its `adv_schema`, its `adv_input_certified` verdict, its stationarity operator, its `adv_conditional_tol` text and the largest `adv_mass_row` over its rows, and its model restrictions, all copied out of the profile's own header, so a reader of the curve alone can tell what it stands on;
- for an `_adv` profile, its derived-state record (`adv_derived_state`), and in the diagnostic mode an `adv_derived_state_refusal` or `adv_derived_state_unknown` line saying that the curve is diagnostic data and not a result;
- the profile's provenance lines;
- the tool's own identity, by git hash and file modification times;
- the census for this line: sampled rows, refused rows, rows above the disk cap, the refusal reasons, the composition breakdown, the eight-point refused-share ladder and its maximum with the b it occurs at, and the sentence saying the share is a contribution and not an uncertainty;
- every `EXHALE_TRANSIT_*` or `TPM_*` override in effect, or a line saying that none is and every window, resolving power and geometry setting is the built-in default or `input.inp`.

When the input states no row validity the census line says so, in the words “the count is UNKNOWN, which is not zero refused”.

6. Validation

Run end-to-end on the converged HD 209458b EXHALE output (output/*_adv.txt), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 Å. With the Huang $J_{Ly\alpha}$ estimate ($\xi = 0.5$, $F_{LyC} \simeq 5.2 \times 10^2$, Γ_2 from $T_* = 6065$ K), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook HD209458b/Halpha_compare_HD209458b.ipynb (and HD189733b/Halpha_compare_HD189733b.ipynb for HD 189733 b).

7. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)

The metal resonance lines are the key NUV/optical diagnostics of metal-enriched escaping atmospheres (Huang et al. 2023). They absorb out of the *ion ground state*, whose excited fine-structure levels are Boltzmann-negligible at $\sim 10^4$ K, so the lower-level density is the ion density itself ($n_{\text{lower}} \simeq n_{\text{ion}}$): no non-LTE population calculation is needed. The ion-density profiles are read from the metal block of Ion_species_adv.txt; a metals-off run (all-zero columns) skips this section automatically.

7.1 Atomic data and implementation

Each species is treated as a **doublet**, with both components summed in a single wavelength window (Table 1; NIST f -values and A -coefficients). The optical depth is computed exactly as for H/He: spherical chords $\sqrt{r^2 - b^2}$ on the impact-parameter grid, LOS velocity $v_{\text{los}} = vx/r$, Voigt profiles via the Faddeeva function (vectorized over wavelength), trapezoidal τ , projected-area disk average, then the instrument convolution and the planet-rotation convolution with the depth-dependent effective radius $R_{\text{eff}} = R_p \sqrt{(h + \delta)/\delta}$ (capped at the impact-parameter boundary), $\delta = (R_p/R_*)^2$. The three doublets are written under the same tpm_<line> naming as the H/He lines (§4.1), with the keys MgII, CaII, NaI.

Table 1: Metal resonance doublets in EXHALE_transit.py (NIST).

species	component	λ_0 [Å]	f	A_{21} [s $^{-1}$]	window / R_{instr}
Mg II	k	2796.352	0.608	2.60×10^8	2790–2810 Å / RES_MGII
	h	2803.531	0.303	2.57×10^8	
Ca II	K	3933.663	0.6267	1.47×10^8	3927–3975 Å / RES_CAII
	H	3968.469	0.3116	1.40×10^8	
Na I	D2	5889.951	0.641	6.16×10^7	5884–5902 Å / RES_NAI
	D1	5895.924	0.320	6.14×10^7	

The resolving powers are the EXHALE_TRANSIT_RES_* values of §4.3: 3×10^4 for Mg II, and the H α value, 1.15×10^5 , for Ca II and Na I. The same list carries the OI 1302 triplet (1302.168, 1304.858, 1306.029 Å, vacuum, NIST), the one entry whose three components absorb out of three different lower levels, the $^3P_{2,1,0}$ fine-structure

populations of `OI_levels_adv.txt`; its band depths are compared with the HD 209458 b measurements in `tpm_OI_band_depths.txt` (user manual, the O I paragraph of the `EXHALE_transit.py` section).

Alongside the spectra, the script prints the validated *single-component* depth table (line-center and 4 Å-band percentages for Mg II 2796, Ca II K, Na I D2), which is the quantity used for the Huang et al. (2023) comparisons; the doublet peak differs from the single-component value only by the partner-line wing (e.g. 3.75 vs. 3.72% for Mg II on WASP-121b). Two conventions matter when comparing with Huang: their table lists the effective transit radius $R_{\text{eff}}/R_{\star} = \sqrt{(R_p/R_{\star})^2 + h}$, not a percent absorption, and Mg II is quoted in a 4 Å NUV bin while the optical lines are line-center (line-center vs. 4 Å differ by $\sim 3\times$ for the optically thick Mg II core).

For a Roche-lobe run the 3-D equipotential reconstruction (`geometry = 'triaxial'`, `roche_recon.py`) replaces the spherical chords by the full Roche geometry with an LOS velocity for each sector $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally locked rotation) and reports the same depth table. Its impact parameters run from 1 to $R_{\text{ib}} = \min(r_N, R_{\star}/R_p)$, the outer edge of the annulus the depth is normalized on, so no chord beyond the stellar radius enters the mean transmission of that annulus.

7.2 Example spectra

Figure 1 shows the three doublet spectra for the WASP-121b worked example (`backup/regression/tpm_wasp/`, Roche Newton solution, metals on). The optically thick cores saturate, so the two doublet members reach similar depths despite the 2:1 oscillator-strength ratio; the rotation convolution mainly shallows and broadens the narrow Na I D cores.

8. Comparison with an observed line profile

The convention below applies to every planet and every line whose model is compared with an observed transit profile (user decision of 2026-10-04, recorded in update log stage 3, section 87, and in the project `CLAUDE.md`). So far it is implemented for one line of one planet, He 10830 of LHS 1140 b: the observed width, the window and the samples are constants of `LHS1140b/he10830_equivalent_width.py`. Another planet or line takes the same steps with its own observation, its own extractor and its own window. `EXHALE_transit.py` itself applies none of them; its curves are the unbroadened, unshifted model.

8.1 The model line is matched to the observed width

A 1-D steady solution produces a line narrower than the measured one (on LHS 1140 b He 10830 about three times: model FWHM about 0.27 Å against 0.841 Å, as recorded in `CLAUDE.md`). The model is therefore convolved with an added Gaussian line-of-sight velocity distribution whose width is fixed by the data, model by model:

1. the excess absorption $E_{\lambda} = (\max T_{\lambda} - T_{\lambda}) / \max T_{\lambda}$ [%] is formed from column 4 of `tpm_He10830.txt`, the curve convolved with the planet's rotation and the instrument's own resolving power (column 3, instrument only, until 2026-10-06; for LHS 1140 b the two equivalent widths differ by $10^{-6} \%$ Å, $v_{\text{rot}} \approx 0.03 \text{ km s}^{-1}$) ($R = 68\,000$ for the WINERED HIRES-Y observation of LHS 1140 b), on its air-wavelength grid (`transit_file_air_excess`);

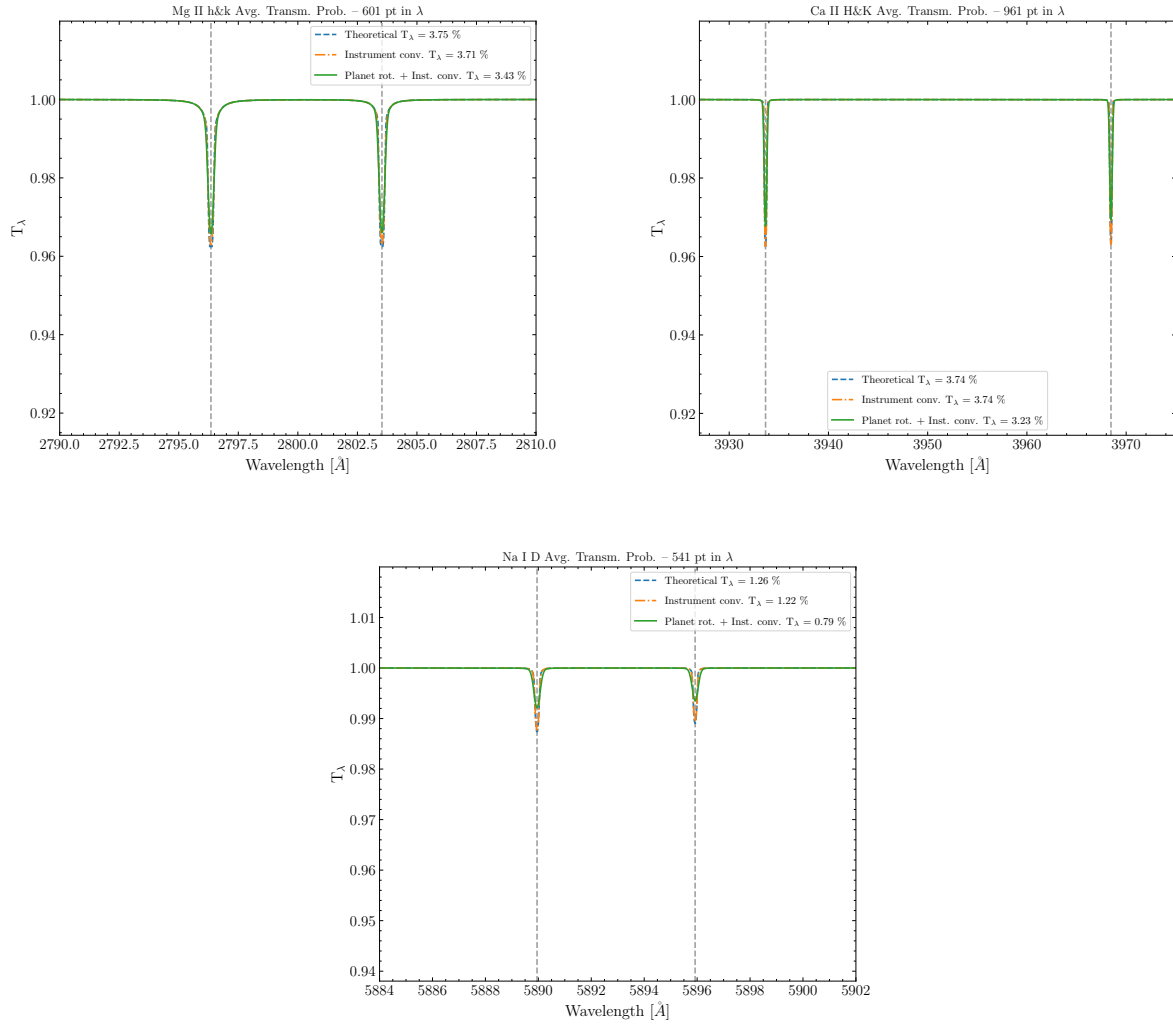


Figure 1: Metal resonance doublet transmission spectra for WASP-121b (backup/regression/tpm_wasp/): Mg II h&k (top left), Ca II H&K (top right), and Na I D (bottom). Each panel shows the theoretical, instrument-convolved, and instrument+rotation-convolved transmission, as for the H/He lines.

2. E_λ is convolved with a Gaussian of FWHM f in velocity, in the planet rest frame and without shift, $\sigma_\lambda = f \times 10830 \text{ \AA} / (2.35482 c)$ (broaden_excess: `scipy.ndimage.gaussian_filter1d`, reflecting boundary, kernel truncated at 4σ , on the median grid spacing). The instrument kernel is already in the curve and Gaussians commute, so this equals broadening before the instrument;
3. depth, width and red/blue ratio of the broadened curve are read with the extractor used on the data (`he_line_metrics.fit_metrics`, air frame); the width is the FWHM of the blended red pair;
4. f is the root of $\text{FWHM}(f) = 0.841 \text{ \AA}$, the same extractor's width of the released spectrum (FWHM_OBS_A), found by Brent's method on $1 \leq f \leq 60 \text{ km s}^{-1}$ with tolerance $10^{-3} \text{ km s}^{-1}$ (`width_matching_fwhm_kms`);
5. the equivalent width compared with the observation is the window equivalent width of the broadened curve (§8.2), and any fit of a planet parameter (He/H, $\dot{M}$, ...) to the observed equivalent width is judged on it.

The kernel stands for a velocity field the 1-D model does not contain. It is not a computed physics term: it carries no energy or momentum and moves no absorber, and every result that uses it reports f and says so. The reflecting boundary conserves the absorption summed over the grid; the window equivalent width falls as f grows because the wings carry absorption out of the window. On the LHS 1140 b solutions f is about 22 km s^{-1} ($21.9\text{--}22.1 \text{ km s}^{-1}$ at He/H $1.8\text{--}3.0$ in update log section 87, and 22.22 km s^{-1} on the 2026-09 solutions of `docs/lhs1140b_exhale_vs_pwinds.tex`, section “The broadening the data demand, measured”; values read from those records, not re-measured here).

Tools. `python3 LHS1140b/he10830_equivalent_width.py --width-matched <tpm_He10830.txt> <output>` [f] writes the broadened excess (`width_matched_product`: columns air wavelength, excess, broadened excess; the header carries f , σ , the criterion, and the EW, depths and width before and after) and prints `fwhm_kms`, `sigma_kms`, `EW_kernel`, `EW_nokernel`, `red`, `blue`, `red_blue`, `fwhm_A`; a given f [km s^{-1}] replaces the root. The evaluation script `LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh <state dir> <work dir> <binary>` runs `Restart intent: stationary evaluate` on a copy of the state (with `conserved_state.txt` staged as the exact code-unit state when present), then `EXHALE_transit.py` at $R = 68000$ on both selections (`transit_adv/`, `transit_solution/`), and writes `EW.txt`. With `EW_NONTHERMAL_KERNEL=width_match` (or a FWHM in km s^{-1}) each product line of `EW.txt` gains `kernel: fwhm_kms= sigma_kms= EW_kernel= EW_nokernel= ...` and `tpm_He10830_broadened.txt` is written; `EW=` stays the unbroadened window EW, and the EW compared with the observation is `EW_kernel`. Unset, nothing changes. `EW.txt` also records the md5 of the consumed files, whether the exact state was staged and read, the evaluate exit status (0; 2, the input state refused; 7, the adv record rejected), the `adv_derived_state` record, and `written_state certified=T|F`, the certification of the state the evaluate wrote, which is the stored pair it loaded. Each product is labeled `result`, `conditional` (an adv record that is unverified), `uncertified` (a solution product on a state not written `certified=T`) or `diagnostic`.

8.2 The equivalent-width operator

The observation and every model go through one operator (`model_equivalent_width`, `observed_equivalent_width`): the excess [%] is interpolated linearly onto the observation's own wavelength

samples inside the window, vacuum wavelengths in the planet rest frame (LHS 1140b: 10832.60–10834.20 Å, 44 samples), and integrated with the trapezoid weights of those samples, in % Å. A model curve in air is converted with the red-pair ratio 10832.057/10829.09114. A curve that does not span the samples, is not finite or is not strictly ordered returns no value instead of an extrapolated one. The observational error is the error of the same linear estimator, $\sigma^2 = \sum_i (w_i \sigma_i)^2$ for independent samples, without covariance or continuum-fit uncertainty, which the released spectrum does not provide; MEASURED with the module on 2026-10-05, $EW_{\text{obs}} = 1.1084 \pm 0.0289$ % Å. The operator applies no line-of-sight velocity and no detector-bin response: models and data are compared unshifted.

8.3 The line-of-sight shift is for figures only

Neither EXHALE_transit.py nor the p-winds comparison spectra apply a bulk line-of-sight velocity, so the model lines sit at the planet rest frame while the measured line is redshifted. In overlay figures the model curves of LHS 1140b are shifted by $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ (the Cherubim et al. 2026 fit value), that is by $\Delta\lambda = (v_{\text{wind}}/c) \times 10830 \text{ Å}$ in air (V_WIND_KMS and dlam_air in LHS1140b/make_memo_figures.py). Every equivalent width, width and depth is computed from the unshifted arrays; the shift enters no number.

9. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter T_star); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting $T_{\text{star}} \leq 0$ reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}}$ from EXHALE is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_ν at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

Reference

- Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144: “H α Absorption in Transiting Exoplanet Atmospheres” (references/Christie_2013ApJ_772_144.pdf).
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150: “A Model of the H α and Na Transmission Spectrum of HD 189733b” (references/Huang_2017_ApJ_851_150.pdf); the $J_{\text{Ly}\alpha}$ estimate is their Eq. (6) and related text.
- Jensen, A. G., et al. 2012, ApJ, 751, 86: transit H α spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).